Detection of Individual Trees in UAV LiDAR Point Clouds Using a Deep Learning Framework Based on Multichannel Representation

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***Abstract*— Individual tree detection is critical for forest investigation and monitoring. Several existing methods have difficulties to detect trees in complex forest environments due to insufficiently mining descriptive features. This study proposes a deep learning (DL) framework based on a designed multichannel information complementarity representation for detecting trees in complex forest using UAV laser scanning point clouds. The proposed method consists of two main stages: ground filtering and tree detection. In the first stage, a modified graph convolution network with a local topological information layer is designed to separate the ground points. Unlike most existing parametric methods, our ground filtering method avoids the optimal parameters selection to adapt to different kinds of environments. For tree detection, a top-down slice (TDS) module is first designed to mine the vertical structure information in a top-down way. Then, a special multichannel representation (MCR) is developed to preserve different distribution patterns of points from complementary perspectives. Finally, a multibranch network (MBNet) is proposed for individual tree detection by fusing multichannel features, which can provide discriminative information for MBNet to detect trees more accurately. MBNet was evaluated on seven forest areas [UAV light detection and ranging (LiDAR) data with the mean size of 14 000 m^2 and point density of 250 points/m^2]. Experimental results showed that the proposed framework achieves excellent performance. Our method obtains promising performance with a mean recall of 89.23% and a mean F1-score of 87.04%.**

***Index Terms*— Deep learning (DL), ground filtering, multichannel representation (MCR), tree detection, UAV light detection and ranging (LiDAR).**

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I. Introduction

FOREST plays an imperative role in the Earth's ecosystem. It has a significant impact on maintaining environmental conditions, such as habitat protection, air quality, and water cycle [1]. These are required for inventorying forest correctly and efficiently. Traditionally, forest inventory needs expensive labor cost, high time consumption [2], and several constraints, such as the weather and field survey conditions [3], [4]. With the progress of light detection and ranging (LiDAR), the UAV laser scanning system is widely used in forest inventory. Due to the ability of capturing 3-D forest structures effectively and accuracy [5], UAV point clouds provide available solutions for numerous forest management tasks, such as estimation of tree species, height, wood volume and crown size, and biomass [6], [7]. Detection of individual trees is important in forest inventory for subsequent estimation of necessary parameters, such as the tree species, height, diameter, crown size, and location.

Generally, tree detection from UAV LiDAR point clouds includes an important preprocessing stage: ground filtering. A suitable filtering algorithm can obtain clean point clouds of nonsurface objects, thus providing good initial data for tree extraction. Several studies have been undertaken to study the problem, such as [8]–[10]. Although these methods achieve good quality results under relatively flat environments, the challenge remains in processing nonflat areas. In this work, we seek the possibility to use a modified graph convolutional networks (GCNs) to improve the filtering performance.

Several methods have been proposed to detect individual trees from UAV LiDAR point clouds. There are three main classes: raster-based methods, point cloud-based methods, and multisource data fusion-based methods. Rasterization is one of the most common ideas. These methods first generate a canopy height model (CHM) by calculating the digital elevation model (DEM) of the Earth surface and canopy surface height. Then, the local maximum is used to determine

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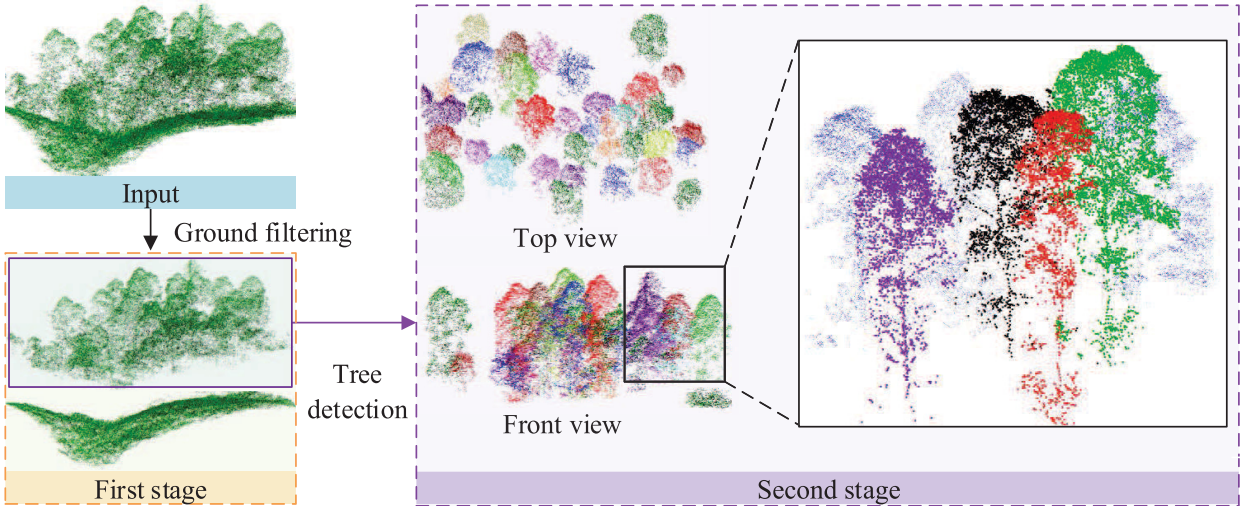


Fig. 1. Two main stages of our framework: ground filtering and tree detection. The ground filtering stage divides the input into off-ground and ground points. The tree detection module takes the off-ground points (purple box in the first stage) as input data and outputs the individual trees.

the potential position of treetop crown, following the pouring algorithm or the region growing (RG) algorithm, to delineate the tree crowns [11], [12]. Some improved CHM-based methods include [13]–[15]. These raster-based methods have achieved significant performance on tree structural attributes determination in the urban simple forest. Since CHM is generated by gridding and interpolation, it inevitably leads to information loss, such as covering some important height information. The determination of tree crowns using region growing or pouring algorithms consumes expensive time. In addition, the rasterized image generated by CHM may not be the most discriminative representation to preserve the spatial relationship among 3-D points, which would have a negative impact on the extraction of tree features. All these shortcomings make raster-based methods difficult to extract trees in complex forests.

Contrary to CHM-based methods, point-based methods analyze directly the 3-D point clouds. The avoidance of generating CHM makes these methods having the main advantage on low information loss as well as mining the spatial structure of forest. In [16], a distance-based segmentation method was presented to generate a tree crown region in a mixed conifer forest. In [17], a voxel-based method is developed to produce a 3-D solid model of a tree for accurate estimation of the volume of the woody material. Lu *et al.* [18] developed a bottom-to-top method based on the intensity and 3-D structure to segment trees in deciduous forests. Vega *et al.* [19] introduced the PTrees, a multiscale dynamic point cloud segmentation method to extract trees. Recently, Ramiya *et al.* [20] presented a supervoxel and local convexity-based method to label trees in urban spaces. Although this method has achieved promising results, the computational performance and ability of processing in complex environments remain a challenge.

Recently, with the development of computer vision and artificial intelligence, deep learning (DL) [21] has been widely applied in image processing. Promising results on various image processing tasks, such as image classification [22], semantic segmentation [23], [24], and object detection [25], [26], have demonstrated its outstanding potential of feature extraction in point clouds. In this work, we attempt to apply the DL approach to improve the descriptiveness of features generated by UAV LiDAR point clouds in complex forests. Our proposed framework consists of two stages: ground filtering and individual tree detection. Specifically, a modified GCN is developed for ground filtering, while a multichannel representation (MCR) is proposed to utilize fully the forest spatial vertical structure information. Then, a multibranch network (MBNet) is designed to mine more discriminative features for tree detection. There are three main contributions of this work.

1) A modified GCNs model is developed for ground filtering. Different from most of the existing parametric filter algorithms, our model is a nonparametric and data-driven method, so it can be applied to various landforms, such as rough slopes, dense vegetation areas, and discontinuous terrains.
2) A top-down slice MCR of forest is developed to reorganize the spatial structure. Compared with traditional representations, such as CHM and point-based representation, MCR can provide more discriminative structure information as well as detailed features to detect trees from complex forests.
3) A multibranch network (MBNet) is proposed to merge multilevel information by concatenating features generated from different branches. Compared with existing approaches, MBNet can mine the distribution pattern to obtain a more discriminative descriptor.

The rest is structured as follows. Section II details the GCN-based filtering method and the MBNet. Experimental results are presented in Section III. Section IV concludes our work.

II. Proposed Method

As shown in Fig. 1, the proposed framework contains two stages: ground filtering and individual tree detection.

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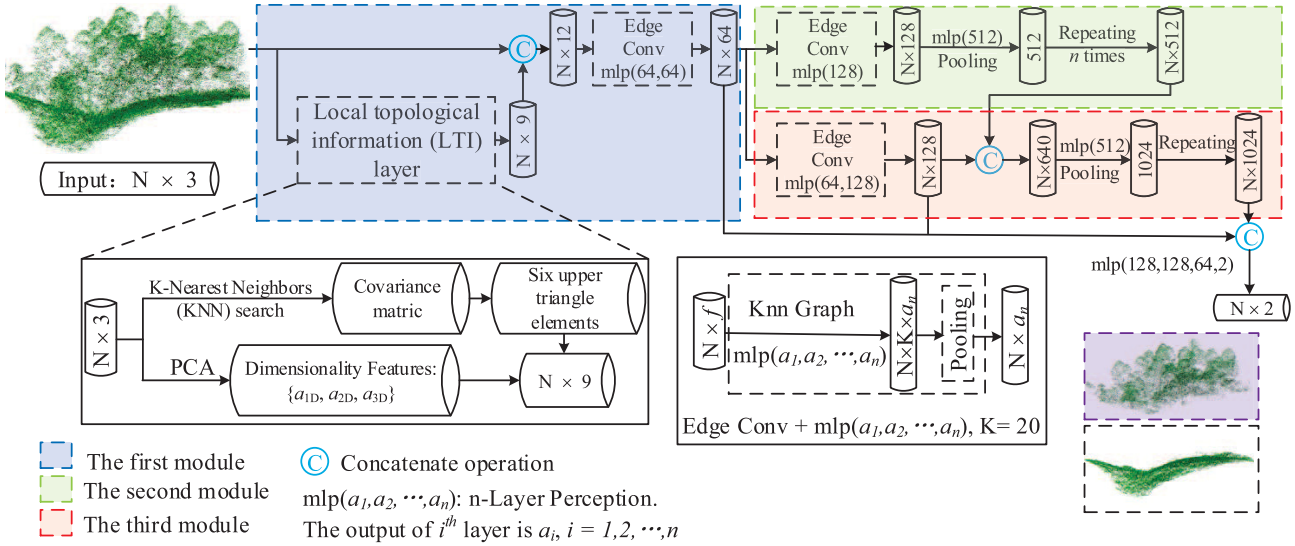


Fig. 2. Flowchart of ground filtering method. It is a modified GCN framework with three modules. The first module is the LTI layer, which provides local features from neighbor points. The second and third modules are the GCN-based networks that designed to describe the global and local features, respectively.

A. Ground Filtering

As mentioned in Section I, most of the existing ground filtering methods can achieve satisfactory results under relatively flat areas, such as the urban environments. However, it is still difficult to filter ground points in complex forests. To improve the performance in abrupt slope, a modified graph convolutional network (GCN) model is developed to learn deep features for ground filtering. As a variant of regular convolutional neural networks (CNNs), GCNs can capture the pairwise dependencies between variables and the information from the graph. Recently, GCNs have achieved huge success in processing graph-structured data, especially in 3-D point clouds, such as FoldingNet [27] and DGCNN [28].

Due to the ability of extracting topological relationships between neighboring points, we introduce GCNs as the basic network to mine the local information to improve the ground filtering performance. More specifically, the ground filtering method consists of three modules (represented by different colors), as shown in Fig. 2. The first module is the local topological information (LTI) layer, which provides a way to mine the local features from neighbor points. The second and third modules are the GCN-based networks that designed to describe the global and local features, respectively.

The LTI layer is the core part, which takes the raw point clouds as input and outputs the shallow features with nine dimensions. More specifically, LTI computes the local covariance information as well as the dimensional features. For each point, 20 neighbor points are searched using the K-nearest neighbors (KNN) search algorithm and the covariance matrix is calculated. Because the covariance matrix is a real symmetric matrix, the six upper triangle elements can be selected as covariance features. In addition, dimensionality features, $(a_{1D}, a_{2\text{-}D}, a_{3\text{-}D})$ [29], [30], are chosen as extra geometric information.

In summary, compared with the original GCN, our modified model enjoys several advantages. First, the LTI layer can provide more detailed information. The local spatial relationship between neighbor points is fully utilized and geometric properties are preserved in the input data. Second, the combination of local and global features enhances the descriptiveness of the modified model. Different from the original GCN, extra global features in our modified model compensate for the deficiencies in the overall characterization. Third, the use of focal loss downweights the loss assigned to well-classified samples (in our work, e.g., a point with a very high or low z value will be a well-classified sample) and pay more attention on the hard samples (e.g., the bushes). Therefore, under the guidance of this loss function, the model can be trained more efficiently in the direction of convergence.

B. Individual Tree Detection

Detecting objects from 3-D point clouds has attracted huge attention. Several works have been developed to undertake this task, including voxel-based methods [31], [32], view-based methods [33]–[35], point-based methods [36], [37] and multirepresentation fusion-based methods [38], [39]. The presence of noise and pseudo outliers caused by mutual occlusion and interpolation in complex forests makes it difficult to apply the voxel- or point-based methods. Possible remedies to avoid noise and outlier effects are to use robust statistical approaches, but usually, it takes more time than the classical approaches for point-based processing [40]. Therefore, in this work, we consider the view-based idea and propose a multibranch network (MBNet). First, different from the typical way of multiview projection, considering the fact that most forests have rich vertical structure information, we design a top-down slice (TDS) module to obtain the multilayer slice representation of forest in a top-down way. Second, to describe the distribution pattern of points contained in different slices, we propose a special MCR module to represent these slices. Finally, based on MCR, a multibranch network (MBNet) module is designed for individual tree detection. Fig. 3 presents the framework of MBNet, which consists of three modules: TDS, MCR, and MBNet.

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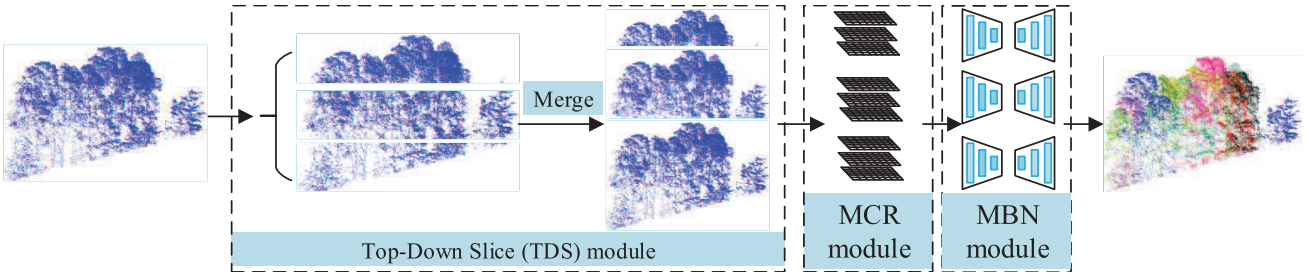


Fig. 3. Flowchart of MBNet. It consists of three modules: TDS, MCR, and MBNet.

1) Top-Down Slice (TDS) Module: After the first stage, we can obtain two separated data subsets: the ground and nonground point subsets. We denote the nonground point set as $P_{\text{non}} = \{p_1, p_2, \ldots, p_n\}$, where n is the number of nonground points. First, the ranges of X, Y, and Z are normalized to [0, 1]. Then, the TDS operation divides P_{non} evenly into different subsets along the z-axis according to a given resolution. TDS is defined as a mapping as follows:

$$\text{TDS}(P_{\text{non}}) = \{L_1, L_2, \ldots, L_K\} \tag{1}$$

where $L_i = \{p | p \in P, \lceil z \times k \rceil = i\}, i = 1, 2, \ldots, K$, K is the number of slices, z is the coordinates of point p along the Z-axis, and $\lceil\ \rceil$ denotes the ceiling function. Finally, as shown in Fig. 3, the merge operation integrates these slices into sequence segments

$$\{S_1, S_2, \ldots, S_K\} \tag{2}$$

$$S_i = S_{i-1} + L_{i-1}, \quad S_1 = L_1, \ i = 2, \ldots, K. \tag{3}$$

Considering the fact that trees in a forest tend to exhibit three distinct vertical distributions: the spherical canopy in the top layer, rod-shaped trunk in the middle layer, and scattered shrub in the bottom layer, we slice the forest into three segments, i.e., $k = 3$. Therefore, the spatial vertical structure of trees would be preserved in these slices.

2) Multichannel Representation Module: The MCR module is designed to obtain the representation of slice generated in the TDS module. Traditional forest representation methods tend to take only the height information. For example, the common CHM-based methods generate CHM by using the height of each point. However, in addition to the height value, several other properties can also provide valuable information in processing complex forest.

Different from the traditional methods, in this work, we consider two additional properties: local height gradient and point density. The local height gradient captures the point distribution change of local position along the vertical direction of the forest. A high gradient means that there is a high possibility of gaps between different trees in the corresponding local region. Therefore, local height gradient is of great importance for determining the canopy. In addition, point density can help to extract treetop more accurately. This is because, according to the plant growth process, there would be more points at the top of tree and its local area. Then, for each slice, we propose a three-channel representation, which consists of density, height, and local height gradient information. More specifically, given arbitrary slice S_i, mapping is designed to map S_i to a three-channel grid image I_i with given resolution r. In this article, we set $r = 512$. Algorithm 1 shows how to generate the mapping.

Algorithm 1 Mapping Slice S_i Into Image I

Input: Slice: $S_i = \{p_j | p_j \in P, \lceil z \times k \rceil = i, j = 1, 2, , n_i\}$; Image resolution r
Output: I: Mapping image
$I = Zeros(r, r, 3)$; $C = Cell(r, r)$;;
// STEP.1: Generating the density channel;
for $j = 1; j \leq n_i$ **do**
  $x = \lceil x_{p_j} \times r \rceil$; $y = \lceil y_{p_j} \times r \rceil$;
  $I(x, y, 1) \leftarrow I(x, y, 1) + 1$; // Density channel;
  $C(x, y) \leftarrow p_j$; // Puting point p_j into $C(x, y)$;
// STEP.2: Generating the Height channel;
for $h = 1; h \leq r$ **do**
  for $t = 1; t \leq r$ **do**
    $I(h, t, 2) \leftarrow |max(z) - min(z)|$; // where z is the z-axis of point p, and $p = (x, y, z) \in C(h, t)$;
// STEP.3: Generating the Height gradient channel;
for $h = 1; h \leq r$ **do**
  for $t = 1; t \leq r$ **do**
    $I(h, t, 3) \leftarrow \sum_{i=-1}^{1} \sum_{j=-1}^{1} \Delta(i, j)$;//where $\Delta(i, j) = |I(h + i, t + j, 2) - I(h, t, 2)|$;
// Normalizing each channel of I to [0, 255];
$I \leftarrow Norm_{[0,255]}(I)$;
return I;

As shown in Fig. 4, each channel preserves the distinctive distribution pattern of trees. Grid image with density channel presents the distribution of points, while height channel captures the location of treetops and describes the edge of canopy. Complementary features contained in these channels can be used to enhance the descriptiveness of the proposed method. Compared with the traditional CHM-based methods, the proposed approach contains more distribution structures, which would contribute to increase the detection performance. Besides, for Algorithm 1, with three loops, its time complexity is $O(n_i) + O(r^2) + O(r^2)$, where n_i is the number of points in slice S_i and r is the given resolution of mapping image. In this article, the image resolution is set as $r = 512$. Therefore, the time complexity of Algorithm 1 can be considered as $O(n_i)$,

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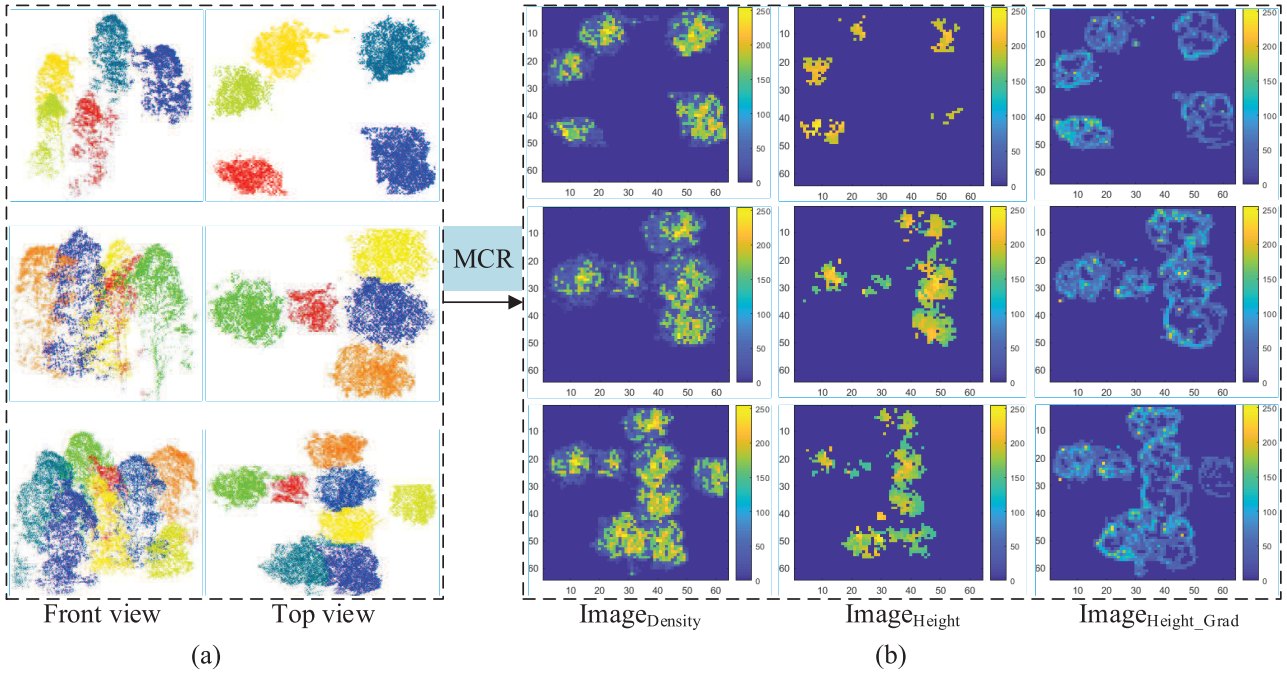


Fig. 4. (a) Raw data and (b) representation generated by the MCR module. Trees in (a) are denoted with different colors manually. Each kind of color represents one individual tree. In (b), Image$_{Density}$, Image$_{Height}$, and Image$_{Height_Grad}$ denote the channel with density, height, and height gradient information, respectively.

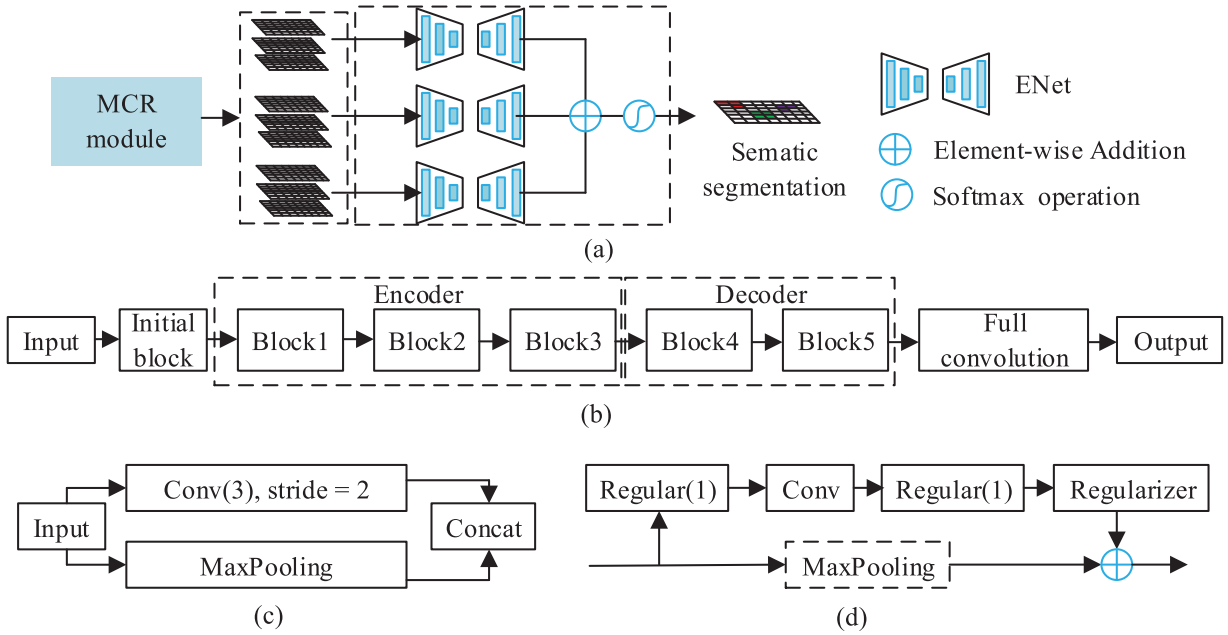


Fig. 5. (a) Flowchart of MBNet module, (b) architecture of ENet, (c) initial block, and (d) bottleneck module in Table I. In the initial block, the size of MaxPooling is 2 × 2 and the stride is 1, while the Conv(3) denotes a 3 × 3 regular convolution with 13 kernels. In the bottleneck module, the main branch consists of three convolutional layers: a 1 × 1 regular convolution that reduces the dimensionality, a main convolutional layer, and a 1 × 1 regular convolution that designs to expand the dimensionality. If there is downsampling in a bottleneck, a MaxPooling operation is added to the main branch. The details are presented in Table I.

i.e., linear in the number of input points. This is acceptable in practical application.

3) Multibranch Network (MBNet) Module: Extracting trees from grid image belongs to the object detection task in image processing. However, since the grid image contains only one type of object (i.e., trees), it is feasible to treat tree extraction as an instance segmentation task. In this work, we use the encoder–decoder (ED) as the basic network. For an ED network, the encoder extracts feature from input data and encodes these features into low-dimensional representation, a feature vector, in a latent space, while the decoder uses the deconvolution operation to recover the input image and predicts the label for each pixel. The ED network has been applied successfully in semantic and instance segmentation, such as U-Net [43], U-Net++ [44], and ENet [45], [46]. The method in [46] is a modified ENet using a well-designed loss

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TABLE I

DETAILS OF ENET. THE INPUT SIZE IS 512 × 512. REGULAR(I) DENOTES A REGULAR CONVOLUTION WITH I × I FILTER. DILATED(I) DENOTES THE DILATED CONVOLUTION [41] WITH I × I FILTER, AND THE ASYMMETRIC(I) DENOTES AN ASYMMETRIC CONVOLUTION [42] WITH I × 1 FILTER AND 1 × I FILTER. FIG. 5 SHOWS THE INITIAL BLOCK AND BOTTLENECK MODULE

Name		Description	Output size		Name	Description	Output size
Initial block		Regular(3)	16×256×256	**Block4**	Bottleneck 4.0	Up sampling + Regular(3)	64×128×128
Block1	Bottleneck 1.0	Down sampling + Regular(3)	64×128×128		Bottleneck 4.1	Regular(3)	
	4×Bottleneck1 .x(x=1,2,3,4)	Regular(3)			Bottleneck 4.2	Regular(3)	
Block2	Bottleneck 2.0	Down sampling + Regular(3)	128×64×64	**Block5**	Bottleneck 5.0	Up sampling + Regular(3)	16×256×256
	Bottleneck 2.1	Regular(3)			Bottleneck 5.1	Regular(3)	
	Bottleneck 2.2	Dilated(1)					
	Bottleneck 2.3	Asymmetric(5)					
	Bottleneck 2.4	Dilated(3)					
	Bottleneck 2.5	Regular(3)		Full convolution			2×512×512
	Bottleneck 2.6	Dilated(7)					
	Bottleneck 2.7	Asymmetric(5)					
	Bottleneck 2.8	Dilated(15)					
Block3	Repeat Block2, without bottleneck 2.0						

TABLE II

MAIN PARAMETERS OF LASER SCANNER

Parameters	Values
Model	Riegl UVX-1
Place of origin	Austria
Size /mm	277×180×125
Weight /kg	3.5
Survey-grade accuracy /mm	10
Laser emission frequency /kHz	550
Scan speed /scans.s^{-1}	200
Field of view /(°)	330

function for the binary instance segmentation and achieves promising performance. Therefore, in our work, we utilize ENet and the loss function in [46] as a basic module and design a multibranch network (MBNet) to learn the semantic features for instance segmentation from three segments generated in the TDS module.

As shown in Fig. 5(a), the MBNet module contains three parallel modified ENets, which have the same network structure. Features generated by each ENet are fused by an adding operation, following a softmax layer. In each ENet, the encoder has three blocks, while the decoder contains two blocks. An initial block is added between input and encoder. A full convolution layer is used to output the final feature maps. Table I presents the details of each block, and Fig. 5(c) and (d) shows the initial block and the bottleneck module, respectively.

It is necessary to point out that the training of MBNet module is performed in a two-step-training way. The first step is the pretrained step. In this stage, each branch is trained individually. Then, we can obtain three pretrained parameter sets. In the second stage, three branches are trained together and the outputs of these branches are fused in an element-wise addition way. The advantage of two-stage-training way is that the model can be trained more steadily and converge faster. This is because these three branches learn their own features from different segments. Training in one step may influence each other and may cause serious oscillations in the optimization process.

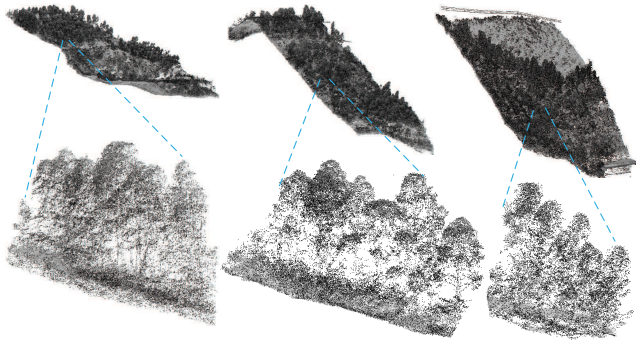


Fig. 6. Samples from study area. These forests are extremely noisy.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Study Area

The study area is the Shaoguan (24°42′N, 113°53′E), located in Guangdong, China. The dataset was acquired with an ALS system, which consists of an unmanned aerial vehicle (UAV) and a lightweight and compact laser scanner. The model of UAV is DJI M600 Pro (SZ DJI Technology Company China). The flying altitude and speed of UAV were up to 150 m and 10 m/s, respectively. The main parameters of the laser scanner are provided in Table II.

Seven areas with complex forests were selected for our experiments. They are in different complex levels. The altitude gap ranges from 106 to 64 m. Fig. 6 shows three samples. Obviously, all these samples contain mixed noise, especially unordered outliers that bring huge difficulty for detecting individual trees. To further analyze these selected areas, much key statistical information is presented in Table III. From Table III, we can see that these areas are of high-density point clouds (approximately 250 points/m^2 in each area). For example, area 1 contains the largest number of points, which is close to 4 500 000. The number of points in area 7 is the lowest, but it is still more than 2 500 000. Besides, it can be inferred from the height information that all these areas are very rough. Especially, the largest altitude gap is 106 m in Area 1. In addition, Fig. 7 presents the height histogram

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TABLE III
KEY STATISTICAL INFORMATION OF SEVEN STUDY AREAS. THEY ARE SORTED ACCORDING TO THE ALTITUDE GAP

	Area1	Area2	Area3	Area4	Area5	Area6	Area7
Size (m^2)	20,231	12,495	11,708	13,168	12,636	15,115	16,110
Point number	4,498,425	4,268,649	3,655,193	3,898,510	3,426,171	3,976,401	2,595,690
Max height (m)	212	212	210	208	201	181	159
Min height (m)	106	107	124	115	123	95	95
Altitude gap (m)	106	105	86	93	78	86	64

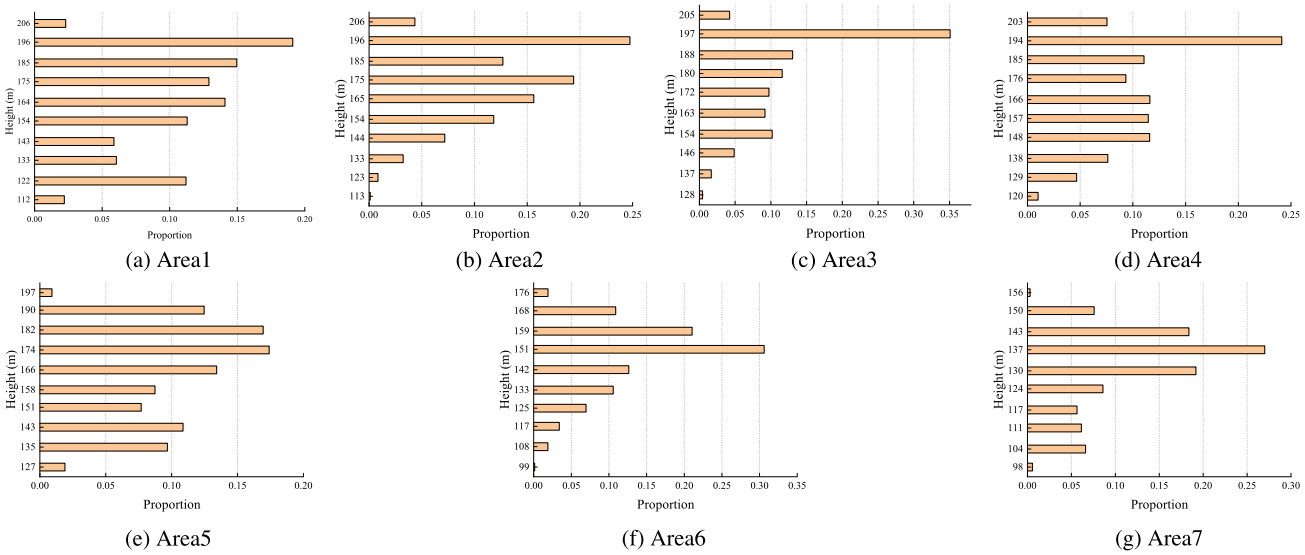


Fig. 7. Height histograms of seven areas. (a)–(g) Histograms of Areas 1–7, respectively.

TABLE IV
DEFINITIONS OF TYPE I ERROR, TYPE II ERROR, TOTAL ERROR, AND KAPPA COEFFICIENT *k*

		Predicted results			
		Ground	Non-ground		
Ground-truth	Ground	a	b	a+b	f=(a+b)/e
	Non-ground	c	d	c+d	g=(c+d)/e
		a+c	b+d	e = a+b+c+d	
Type I error		b/(a+b)			
Type II error		c/(c+d)			
Total error		(b+c)/e			
k	[(a+d)/e - p]/ (1 - p), where p = f×(a+c)/e + g×(b+d)/e, a,b,c and d are the true positive, false negative, false positive and true negative, respectively.				

for each area, which shows that the terrains are diverse and no-flat. Therefore, these selected areas are suitable for our study. It needs to point out that the ground truth for ground filtering algorithm and individual tree detection in this work are obtained by careful manual classification. Specially, they are independently labeled and verified by three people to make the annotation as accurate as possible.

Several experiments are conducted to evaluate the proposed method in the following subsections, including tree detection using MBNet, efficiency testing, and model design analysis.

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TABLE V
OPTIMAL PARAMETERS FOR CSF IN SEVEN AREAS

	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6	Area 7
Cloth resolution (m)	2.04	9.11	5.88	7.75	2.38	5.48	4.04
Max iterations	200	750	500	800	500	300	450
Classification threshold (m)	4.57	11.85	8.93	8.18	4.04	11.33	7.46

TABLE VI
COMPARING GROUND FILTERING RESULTS GENERATED BY DIFFERENT METHODS. I, II, T., AND k DENOTE TYPE I ERROR, TYPE II ERROR, TOTAL ERROR, AND KAPPA COEFFICIENT, RESPECTIVELY

Area	Metric	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6	Area 7	Means	Max	Min
CSF [47]	I (%)	2.49	2.04	**2.01**	**4.57**	3.31	6.53	**3.54**	**3.49**	-	2.01
	II (%)	9.34	9.11	8.66	11.85	14.42	16.6	12.25	11.74	-	8.66
	T. (%)	6.00	5.88	5.03	8.93	8.45	12.28	8.53	7.87	-	5.03
	k	87.52	87.75	89.12	81.86	82.3	74.75	81.46	83.53	89.12	-
PointNet [36]	I (%)	4.28	2.58	4.32	9.18	2.97	8.98	8.83	5.88	-	2.58
	II (%)	**9.14**	6.22	4.06	8.51	15.41	11.26	5.95	**8.65**	-	4.06
	T. (%)	5.92	4.35	4.24	8.96	7.68	10.75	7.15	7.00	-	4.24
	k	87.21	90.76	90.32	81.64	82.48	77.73	84.29	84.91	90.76	-
DGCNN [28]	I (%)	4.12	2.00	3.20	7.68	2.82	8.97	8.20	5.28	-	**2.00**
	II (%)	9.62	5.92	4.02	**8.34**	**13.31**	**11.25**	**5.81**	8.32	-	4.02
	T. (%)	6.01	3.91	**3.43**	8.15	6.82	10.73	6.78	6.54	-	3.43
	k	86.99	91.69	**92.13**	83.30	84.49	77.75	85.09	85.92	92.13	-
Ours	I (%)	**2.39**	**1.85**	4.23	6.86	**2.09**	**5.66**	7.05	4.30	-	1.85
	II (%)	10.49	**5.85**	**3.71**	9.09	14.76	15.15	7.03	9.44	-	**5.85**
	T. (%)	**5.41**	**3.77**	4.00	**7.91**	**6.79**	**10.59**	**6.80**	**6.46**	**-**	**3.77**
	k	**88.14**	**91.98**	90.85	**83.65**	**84.47**	**77.84**	**85.13**	**86.00**	**91.98**	**-**

B. Ground Filtering

In this work, we conducted experiments on the above datasets and compared our method with the previous ground filtering algorithm, including the DL-based methods, PointNet [36] and DGCNN [28], classic method, and cloth simulation filtering (CSF) proposed in [47]. CSF uses the cloth simulation algorithm to achieve the state-of-the-art results and is available in the open-source CloudCompare (http://www.cloudcompare.org/).

We used the metrics provided in [48] and [49] to measure the performance. As shown in Table IV, the work [48] proposed three metrics for quantitative analysis. Type I error measures the rate of ground points mislabeled as nonground points, whereas Type II error represents the percentage of nonground points mislabeled as ground points. The total error shows the rate of all mislabeled points. Besides, Cohen's kappa coefficient (k) [49] is widely used in most of filtering algorithms [50]–[52]. It measures the inter-ratio agreement more robustly than a percentage. Since the CSF is a parameter method, we evaluated the performance of CSF on each area using the grid search method to obtain the optimal parameters. Table V shows the optimal parameters for seven areas. It needs to point out that, as ours is a supervised method, training data are required. Therefore, in our experiments, when an area is used as testing data, other six areas are considered as training data. In addition, our method was trained with TensorFlow on a NVIDIA Tesla P100 GPU. The batch size was set to 8 and the initial learning was 0.001. When training the model, we used adaptive moment estimation (Adam) with a momentum of 0.9. The number of epochs was 50.

Table VI presents the compared results. Obviously, our method achieves excellent performance. More specifically, our method obtains the best results on Area 2. Besides, in Areas 1, 4, 5, 6, and 7, our method also achieves competitive results.

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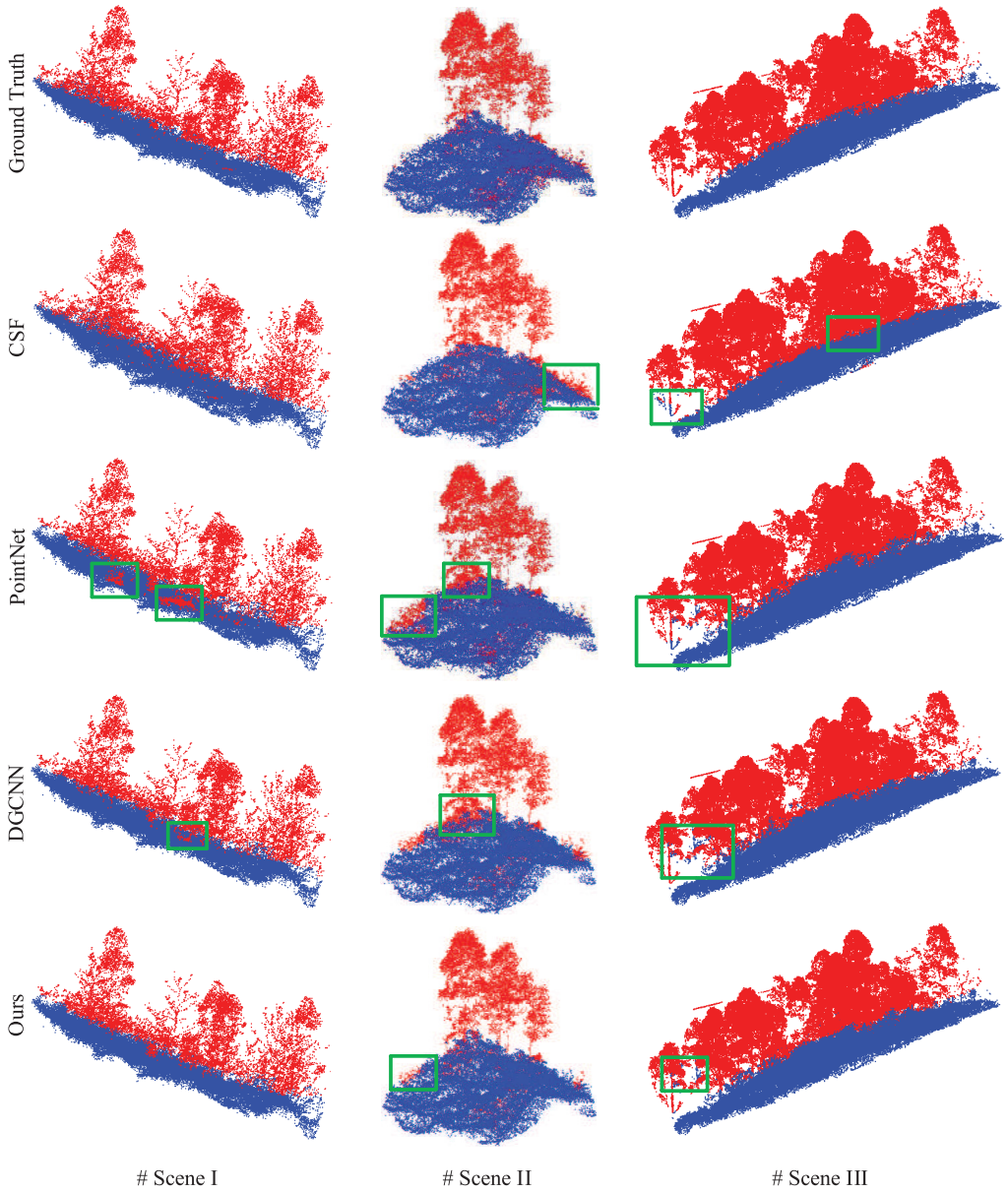


Fig. 8. Results of ground filtering generated by different methods on several scenes. The ground and nonground points are marked by blue and red colors, respectively. The areas marked by the green boxes are areas with more mislabeling error points.

In addition, compared to CSF and PointNet, our method increases the performance by 2.47% and 1.08% in overall average k coefficient, respectively. These results mean that our method can filter the ground points more precisely. On the overall average total error and Type II error, our method also has the obvious advantage, with a significant reduction compared with other three methods. This means that our method can reject object points more effectively and has a smaller proportion of all error points.

The reason for the better performance of our method is that the local graph structure used in the ground filtering framework can preserve the relationship among neighbor points. Compared with PointNet and DGCNN, the local features of ground points are captured by our method more effectively. Besides, compared with CSF, as a nonparametric method, our method does not need to find the optimal parameters.

To further present the comparative results, we visualized the ground and nonground points for some areas. As shown in Fig. 8, our method achieves excellent performance in preserving ground points. Note that filtering ground points in steep areas is another challenge [10]. Our method also labels the ground points as nonground points in some areas, just as shown in scenes II and III. However, the number of mislabeling points is very low and acceptable in practice.

C. Tree Detection Using MBNet

In our work, tree detection contains three steps, including building the synthetic dataset, training MBNet on synthetic dataset, and testing MBNet on both synthetic and real datasets.

1) Synthetic Dataset Generation: Generating synthetic data is of great importance for the proposed method. It is known

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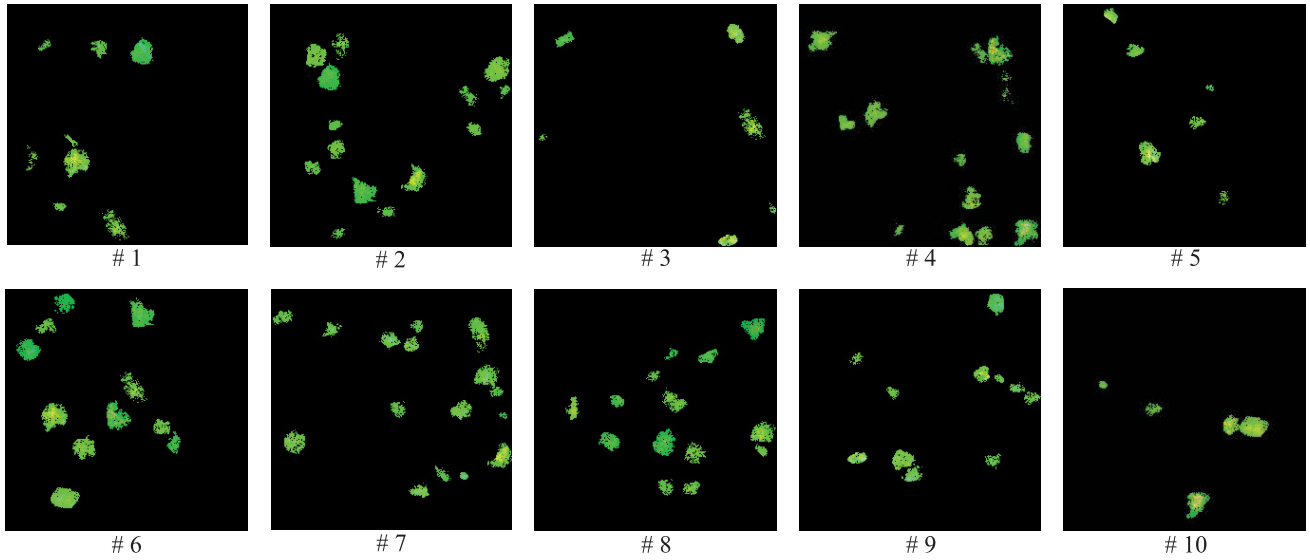


Fig. 9. Several samples from the synthetic dataset.

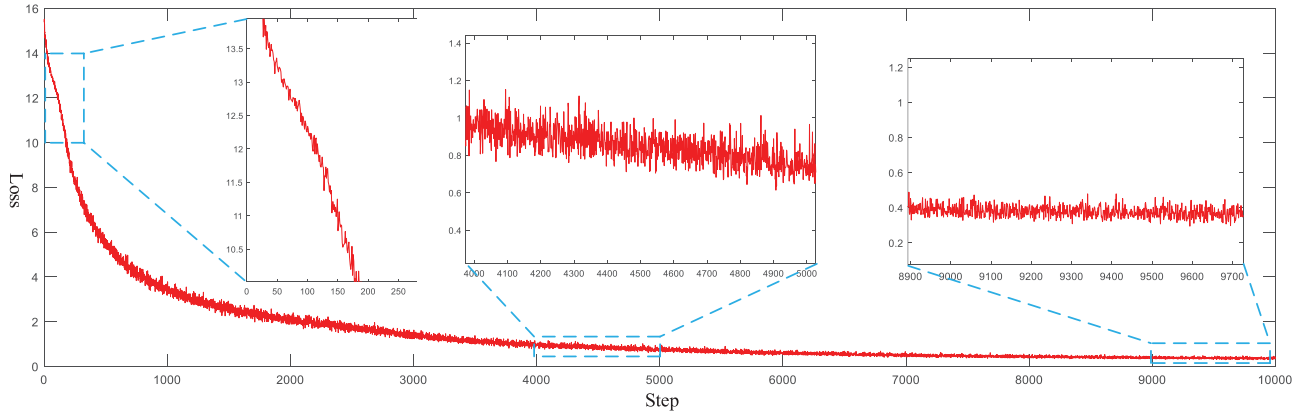


Fig. 10. Curve of training error. The loss decreases rapidly in the early stage and tends to be stable in the later stage. Starting from around 9000 steps, the model converges.

TABLE VII
DETAILS OF THE SYNTHETIC DATASET

	Training sample number	Testing sample number	Mean point number	Mean tree number
Synthetic dataset	1,200	200	22,309/sample	12/sample

that a supervised method based on DL requires a large number of labeled training samples. For example, in the image processing field, ImageNet [53], a huge dataset, provides a variety of labeled images to improve the performance of image processing methods. Similarly, in our work, we designed a simple but effective method to generate a large number of synthetic samples. Specifically, after ground filtering, individual trees were first extracted manually from the nonground points. Then, part of these individual trees was selected randomly and put together to form a sample. Note that, to better simulate the real scene, the location of each tree is also random. Finally, trees in this sample are projected into a 2-D grid image using Algorithm 1. A large number of training samples can be obtained by repeating this way of random sampling. Table VII describes the detail of synthetic dataset and Fig. 9 shows several samples.

2) Training MBNet on Synthetic Dataset: The proposed model was trained on the above synthetic dataset. We trained the model with Tensorflow on a NVIDIA Tesla P100 and the initial learning rate is set to 0.001 and decreased by half in every 20 epochs. The batch size is 32. The pretrained stage has the same hyperparameters. Fig. 10 shows the curve of training error. It was found that in the first 1000 steps, the error decreased significantly, which means that through the previous training, the model has obtained the optimal direction of descent. Then, from 1000 to 7000 steps, the deceleration

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TABLE VIII
COMPARATIVE RESULTS OF TREE DETECTION GENERATED BY DIFFERENT METHODS

	RG-CHM			Mask-RCNN [54]			Our method		
	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
Area 1	**97.79**	26.25	41.38	77.93	45.69	57.60	94.44	**83.51**	**88.64**
Area 2	85.86	**86.31**	86.08	74.69	46.11	57.01	**97.14**	77.95	**86.49**
Area 3	77.36	58.82	66.83	73.06	53.86	62.00	**93.09**	**89.68**	**91.35**
Area 4	**77.78**	16.67	27.45	68.69	70.11	69.39	**77.78**	**95.83**	**85.87**
Area 5	75.09	8.33	15.00	68.69	56.52	62.01	**80.00**	**95.24**	**86.96**
Area 6	**82.36**	61.46	70.39	71.96	44.51	55.00	74.34	**91.96**	**82.22**
Area 7	70.51	80.95	75.37	75.24	66.10	70.37	**85.19**	**90.48**	**87.75**
Means	80.96	48.39	54.64	72.89	54.70	69.91	**85.99**	**89.23**	**87.04**

of error rate gradually slows down. This means that the model is steadily moving toward the direction of convergence. Finally, starting from around 9000 steps, the error tends to be stable, which means that the model has converged. Therefore, considering that in this work, the number of training samples is 1200 and the batch size is 32, the number of epochs is set to 250.

3) Testing MBNet: We evaluated the proposed detection approach on real datasets, i.e., seven areas. The Mask-RCNN [54] and a baseline, classic CHM- and RG-based algorithm (denoted as RG-CHM) were used for comparing with the proposed method. Considering the complexity of forest, we studied the most important aspect of performance for each sample, the number of trees. The following three metrics are used to measure the performance:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$\text{F}_1 \text{ score} = \frac{2 \times \text{Precesion} \times \text{Recall}}{\text{Precesion} + \text{Recall}} \tag{6}$$

where TP, FP, and FN are the number of true positives, false positives, and false negatives, respectively.

Table VIII presents the performance of different methods. As shown in Table VIII, by achieving mean precision of 85.99%, mean recall of 89.23%, and mean F1-score of 87.04%, our method outperforms the baseline and Mask-RCNN. Furthermore, in each area, the F1-score of our method is higher than that of other two methods.

The better results achieved by our method mainly because of two reasons. First, the multibranch representation of forests preserves the distribution patterns of tree points with different heights. Compared to the CHM representation used in the RG-CHM method, the MCR contains rich hierarchical structure information that can improve the descriptiveness of our method significantly. Second, compared to Mask-RCNN, our designed multibranch network first takes MCR as input, which would provide rich vertical distribution information at different heights. Then, the fusion module used in our network allows the information contained in each 2-D grid image to be fused together to complement each other, which would enhance the ability of feature extraction.

TABLE IX
INFERENCE TIME OF DIFFERENT METHODS ON REAL DATASET OF SEVEN AREAS IN SECONDS (s), WHERE "Ai" MEANS "Area_*i*"

Method	Mean	A1	A2	A3	A4	A5	A6	A7
RG-CHM	31.01	45.82	20.93	52.37	3.94	60.72	26.46	6.89
Mask-RCNN	0.61	1.29	0.65	0.72	0.16	0.54	0.66	0.26
Ours	4.77	7.98	4.98	6.43	1.42	4.91	5.62	2.06

Fig. 11 shows several detection results from the 2-D grid image generated by different methods. Obviously, compared with RG-CHM and Mask-RCNN, our method has a more significant advantage over them. Furthermore, our method has a better balance between the recall and the precision. For each sample, the area of crown and the number of trees predicted by our method are more accurate than RG-CHM and Mask-RCNN.

D. Efficiency Testing

We first evaluate the time efficiency of different methods. Specifically, we compare our method with RG-CHM and Mask-RCNN by calculating the inference time on real dataset of seven areas. Table IX presents the comparison results. Obviously, our method achieves better performance than RG-CHM on each area. More specifically, RG-CHM is about six times slower than our method in terms of average time. The reason for the high efficiency of our method mainly lies in the acceleration of GPU, which can execute the network quickly. However, compared with the DL-based method, the Mask-RCNN, our method has a high time consumption. Mask-RCNN is more than seven times faster than our method in

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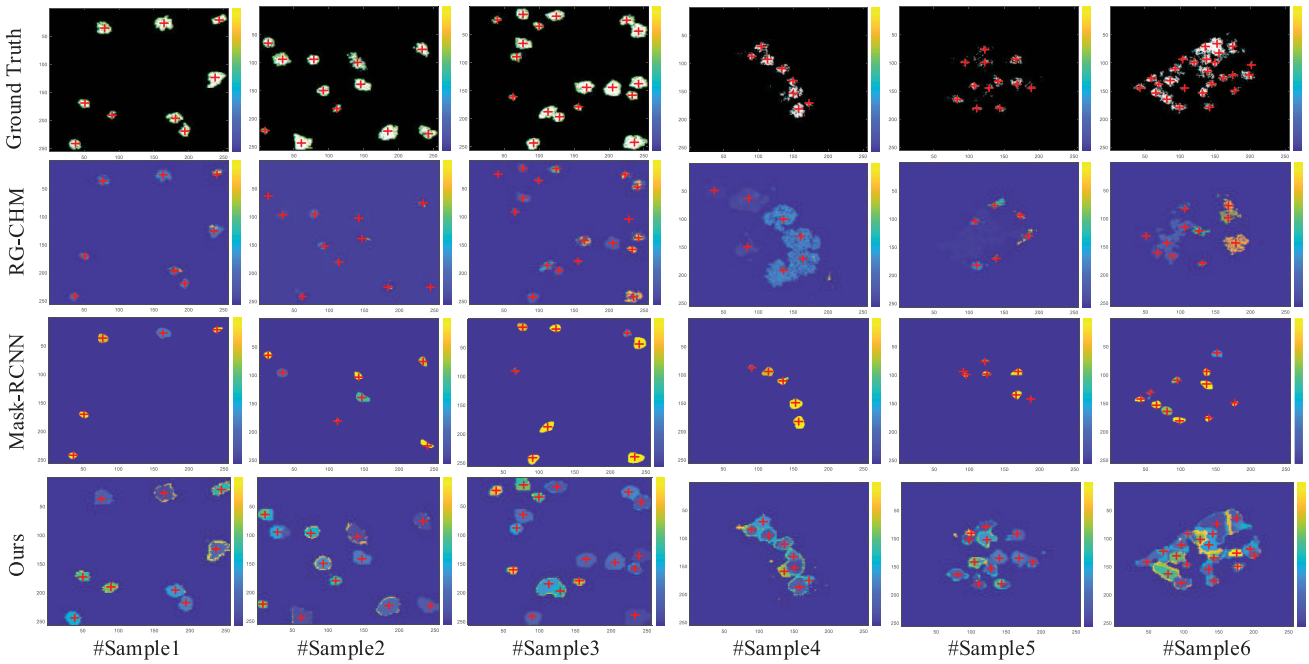


Fig. 11. Tree detection results generated by different methods on several samples. The first three columns are the results from the synthetic dataset, while the last three columns are the results generated from the real dataset. "+" with red color denotes an individual tree.

TABLE X
COMPARISON OF DIFFERENT REPRESENTATIONS AND OUR METHOD

	Real dataset		
	Precision (%)	Recall (%)	F1 score (%)
Net_D	93.85	70.86	80.06
Net_H	**98.06**	46.37	61.75
Net_{HG}	96.54	61.83	73.73
Ours	85.99	**89.23**	**87.04**

terms of average time. This is because our method takes the multibranch network and needs to fuse the features generated by multiply branches, which would bring lower time efficiency.

Second, we calculate the parameters of the proposed method. The number of parameters is about 1.11 M, where M stands for million. Besides, we report storage required to save model parameters in half-precision floating-point format and the required space is about 2.1 MB, which means that our model has a lightweight size and can be applied for real project in the forest inventory.

E. Model Design Analysis

As discussed in Section III, our designed framework consists of three representations: the density-, height-, and height gradient-based grid images. To analyze our model, we evaluated the effects of each representation.

Net_D, Net_H, and Net_{HG} are used to denote the proposed method only with the density-, height-, and height gradient-based grid images, respectively. We compared our method with these three baselines. As shown in Table X, our method achieves the best performance in terms of recall and F1 score. Specifically, our method significantly outperforms other three baselines by a margin on the real dataset. Compared with Net_D, our method achieves significant improvements of 18.37% and 6.98% of recall and F1-score, respectively.

The above results show that our method has a better balance between precision and recall. The reason lies in the multibranch fusion strategy. The three representations used in the designed framework preserve different information in three aspects, which can provide rich complementary structure features to enhance the descriptiveness. However, it needs to point out that in terms of precision, our method is worse than Net_D, Net_H, and Net_{HG}. Besides, we can observe that Net_H achieves the best performance in terms of precision. This is because for Net_H, comparing our method, it takes only the height information as its representation, and the tree points are extracted more strictly. This means that Net_H would obtain a lower FP. According to the definition of precision, Net_H can obtain a higher precision.

To further investigate the effects of each representation, we evaluate the performance of different representations in each area. Fig. 12 provides the details for the performance of different representations in seven areas. More specifically, the precision/recall/F1 score in Table X is the average value of Fig. 12(a)–(c) in seven areas. As shown in Fig. 12(b) and (c), we observe that our method obtains the highest recall and F1 values, respectively, in each area. This shows that our method is more descriptive than three baselines. In addition, among three baselines, Net_D performs well in terms of recall, while NetH achieves better performance in precision, whereas Net_{HG} produces the worst results. This is consistent with the expectation of these representations. Representations of Net_D and Net_H are designed to capture the distribution patterns of

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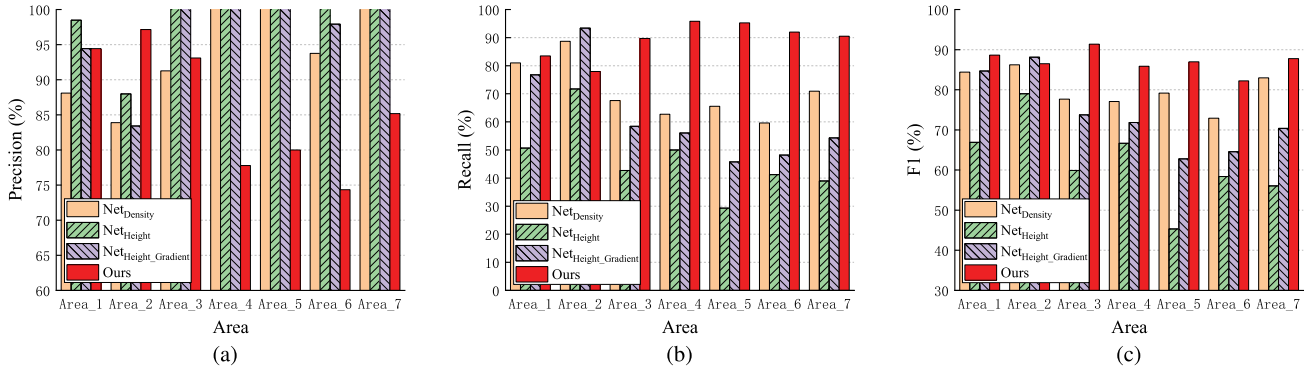


Fig. 12. Performance of different representations on seven areas. The metrics are (a) precision, (b) recall, and (c) F1 score.

density and height of points, respectively, while that of Net_{HG} encodes the gradient of height, which is mainly expected to provide additional supplementary information. Therefore, these representations make their own contributions to the designed framework.

IV. Conclusion

This work aims to extract more discriminative features for ground filtering and individual tree detection in complex forest. At the ground filtering stage, a local topological-based GCN is designed to mine the relationship among neighbor points to improve the ground filtering performance. As a data-driven approach, the modified GCN avoids the parameter selection problem associated with most of the existing parametric methods. Compared to CSF and PointNet, our method increases the performance by 2.47% and 1.08% in overall average k coefficient, respectively. Compared to DGCNN, our method also achieves better results. At the detection stage, unlike most of the existing methods mainly focused on the height information of forest, we first developed an MCR to preserve three kinds of distribution patterns of points in three complementary perspectives: density, height, and height gradient of points. Second, based on MCR, a multibranch network (MBNet) is designed to fuse the hierarchical structure features to detect trees. The proposed MBNet presents a promising way to apply the DL to extract deep features from complex forest accurately. Experimental results show the superiority of the proposed architecture.

Limitation and Future Work: First, as the description in part II, our method mainly focuses on the point clouds with obvious differences in spatial distribution. More specifically, our method mainly deals with the trees with larger crown coverage and smaller trunk space. Therefore, generally speaking, the proposed method is more suitable for the large broad-leaved forest, such as the study area of this work, the Shaoguan, located in Guangdong, China, which belongs to the subtropical monsoon climate zone. For the common trees in other areas, such as coniferous forest in the north, our method will be limited due to the small difference in vertical spatial distribution. Second, since our method is designed mainly for individual tree detection, it cannot be used directly for more detailed analysis. However, the results obtained by our method can be used for downstream tasks. For example, building on our method, several parameters, such as the tree height and crown size, can be obtained by designing inverse mapping from 2-D image to 3-D point cloud, which would be an important topic in our future work. Third, it also needs to point out that the point cloud quality has an important influence on the performance of tree detection. For example, the point cloud quality, especially the point cloud density, has a significant impact on the second module, i.e., MCR module. More specifically, for the computation of density channel in Algorithm 1, fixing the image resolution r, if the point cloud density is low, the mapping image will be difficult to accurately preserve the distribution pattern of the point cloud. In future work, we will explore an adaptive resolution setting method so that the mapping image resolution r can adapt to the point cloud density. In addition, as shown in Table IX, compared with Mask-RCNN, because of the use of multiply branches, our method has lower time efficiency. Besides, as a data-driven approach, the performance of our method would be dependent heavily on the requirement of a large number of labeled samples, which would consume a lot of time and labor costs. Therefore, as further work, generating more samples from real forest environments and investigating unsupervised methods, such as domain adaptation, will be focused.

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